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PBL PHASE-1 REPORT

PROJECT TITLE

**DBOS-Based Resource Management Platform
with EV Charging as a Real-World Workload**

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PROJECT AND TEAM INFORMATION

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DBOS-Based Resource Management Platform with EV Charging as a Real-World Workload

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Abstract

- The proposed project presents a **DBOS-Based Resource Management Platform** that combines Operating System resource management concepts with Database Management System capabilities.
- The platform uses a database as a central source of system state and resource information.
- **EV charging** is selected as a real-world workload because charging stations contain limited resources such as charging points, power capacity, and charging time.
- Multiple users may request charging resources simultaneously, creating resource allocation, scheduling, synchronization, and concurrency problems.
- The proposed system aims to manage these requests using database-backed resource allocation and OS-inspired scheduling.
- The system will maintain information about users, charging stations, charging requests, reservations, resource allocation, transactions, and system logs.
- The project demonstrates how OS and DBMS concepts can work together to create a reliable resource management platform.

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1. Introduction

1.1 Background

The project is based on the concept of a **Database-Oriented Operating System (DBOS)**.

The DBOS approach proposes using a high-performance database as a foundation for operating-system state and services. The DBOS research direction explores storing OS state in a distributed database and using database transactions for system operations. [1]

The proposed project applies these concepts to a resource management scenario.

1.2 Project Idea

The project focuses on:

- Combining OS resource management with DBMS state management.
- Treating resource requests as system tasks.
- Maintaining resource state using a database.
- Scheduling multiple resource requests.
- Preventing resource allocation conflicts.
- Maintaining transaction history.
- Using EV charging as the practical workload.

2. Problem Statement

- EV charging stations have a limited number of charging resources.
- Multiple users may request charging simultaneously.
- A charging point cannot be allocated to multiple users at the same time.
- Requests must be organized and scheduled when resources are busy.
- Resource availability must be updated immediately after allocation or release.
- Concurrent requests can create race conditions if not handled properly.
- Failed transactions may result in inconsistent resource status.
- Historical resource usage must be stored for monitoring and analysis.
- Therefore, an integrated resource management mechanism is required to manage allocation, scheduling, synchronization and database state consistently.

3. Motivation

The project is motivated by the following points:

- Increasing use of electric vehicles requires efficient charging infrastructure.
- Charging stations have limited physical and electrical resources.
- Efficient resource allocation can reduce unnecessary waiting.
- Database transactions provide mechanisms for consistency and reliable state management.
- Operating System concepts provide scheduling and resource management techniques.
- DBOS provides an opportunity to combine these capabilities.
- The project provides a practical example of OS and DBMS integration.

4. Objectives

The major objectives are:

1. To design a DBOS-inspired resource management platform.
2. To represent system resources using database-backed state.
3. To manage EV charging requests efficiently.
4. To implement resource allocation and release mechanisms.
5. To design a scheduling mechanism for competing requests.
6. To handle concurrent requests safely.
7. To maintain reliable transaction records.
8. To prevent duplicate resource allocation.
9. To maintain system activity logs.
10. To demonstrate the integration of OS and DBMS concepts.

5. Project Scope

5.1 Included Scope

- User registration and identification.
- Charging station registration.

- Charging point management.
- EV charging request submission.
- Resource availability checking.
- Request scheduling.
- Resource reservation.
- Resource allocation.
- Resource release.
- Transaction management.
- Database logging.
- Monitoring and reporting.

5.2 Possible Applications

The same resource management approach can be extended to:

- Cloud computing resources.
- CPU scheduling.
- Shared laboratory equipment.
- Smart parking systems.
- Battery swapping stations.
- Smart energy management.
- Server resource allocation.

6. Existing System

Traditional resource management architectures generally separate operating-system services and database management.

Major limitations include:

- System state may be maintained separately from application data.
- Synchronizing state between multiple components can increase complexity.
- Resource allocation conflicts may occur under concurrent requests.
- Monitoring historical resource state may require additional systems.
- Transactional guarantees may not automatically cover all resource management operations.

- Different services may implement their own state-management mechanisms.

7. Proposed System

The proposed system integrates OS and DBMS concepts.

- The database stores resource and system state.
- Resource requests are represented as tasks.
- A scheduler determines the execution order.
- A resource manager checks resource availability.
- Synchronization mechanisms prevent conflicting allocation.
- Transactions maintain database consistency.
- Logs maintain a history of system activities.
- EV charging provides a practical workload for demonstrating the system.

8. Operating System Concepts Used

8.1 Resource Management

- Charging points are treated as shared system resources.
- The resource manager maintains their availability.
- Resources are allocated to eligible requests.
- Resources are released after completion.

8.2 Scheduling

- Charging requests are treated as tasks.
- The scheduler determines which request should execute first.
- Possible scheduling policies include:
 - First Come First Serve (FCFS).
 - Priority Scheduling.
 - Shortest Job First (SJF).
 - Time-based scheduling.

8.3 Synchronization

- Prevents two users from acquiring the same charging point.
- Reduces race conditions.
- Maintains correct resource state.
- Coordinates concurrent resource requests.

8.4 Concurrency

- Multiple EV requests may arrive simultaneously.
- Concurrent requests must be processed safely.
- Database transactions and locking mechanisms can be used to maintain consistency.

9. DBMS Concepts Used

9.1 Database State Management

The database can maintain:

- User information.
- Charging station information.
- Charging point status.
- EV charging requests.
- Reservations.
- Resource allocations.
- Transaction records.
- System logs.

9.2 Transactions

A charging allocation can be represented as a transaction:

1. Check charging point availability.
2. Lock or reserve the selected resource.
3. Create the allocation record.

4. Update resource status.
5. Commit the transaction.
6. Release the resource after completion.

If an operation fails, rollback can be used to restore a consistent state.

9.3 Concurrency Control

Concurrency control is required because:

- Multiple users can request the same resource.
- Resource status can change rapidly.
- Simultaneous updates can produce inconsistent results.
- Database locking and transaction isolation can reduce these problems.

10. Proposed Database Design

The proposed database may contain the following tables:

Table	Purpose
Users	Stores user identification and account information.
Charging Stations	Stores charging station details.
Charging Points	Stores individual charging point status and availability.
Charging Requests	Stores EV charging requests.
Reservations	Stores resource reservation information.
Allocations	Stores resource allocation details.
Transactions	Stores transaction history.
Logs	Stores system activities and events.

11. System Architecture

The proposed architecture combines the application layer, resource-management layer, scheduling layer, synchronization layer and database layer.

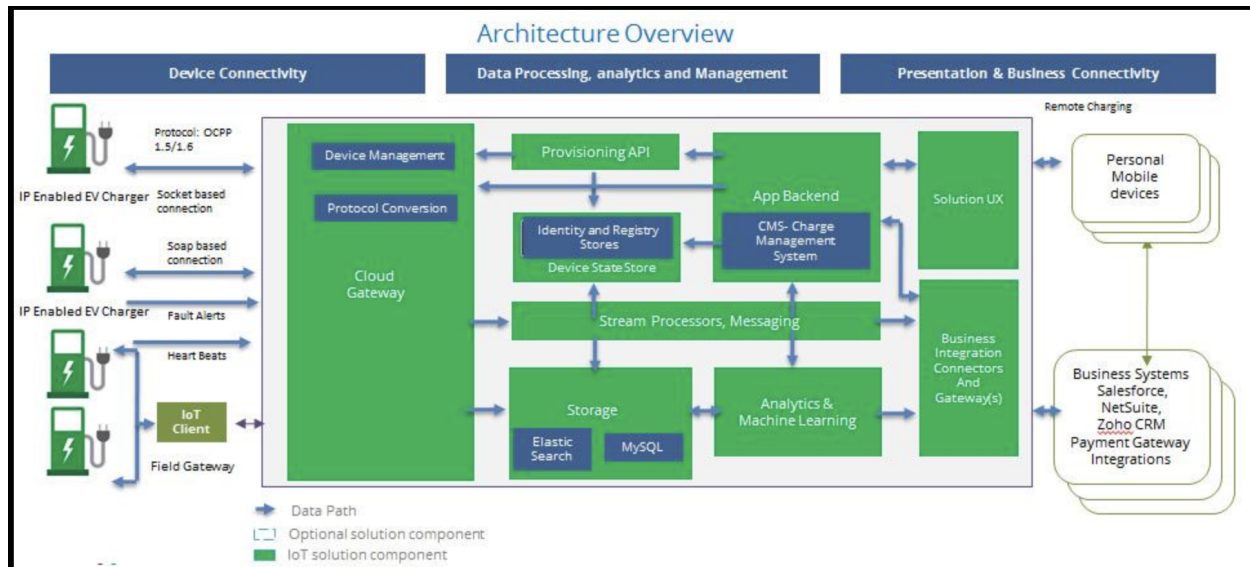


Figure 1: Proposed Architecture of the DBOS-Based Resource Management Platform

12. Architecture Components

1. User / Client Layer

- Allows users to submit EV charging requests.
- Provides information about charging status.

2. Application Layer

- Receives user requests.
- Validates request information.
- Sends valid requests to the resource manager.

3. DBOS Resource Manager

- Maintains resource information.
- Checks availability.
- Coordinates resource allocation.

4. Scheduler

- Maintains waiting requests.
- Selects the next request for execution.
- Applies the selected scheduling policy.

5. Synchronization and Concurrency Control

- Prevents conflicting resource allocation.

- Controls simultaneous access.

6. Database

- Stores system state.
- Stores transactions.
- Stores resource allocation history.

7. Monitoring and Logging

- Records system events.
- Supports debugging.
- Supports resource usage analysis.

13. Working of the Proposed System

The system workflow is:

1. A user submits an EV charging request.
2. The application validates the request.
3. The request is stored in the database.
4. The resource manager checks charging point availability.
5. If a resource is available, the scheduler selects the appropriate resource.
6. The system reserves the resource using a transaction.
7. Synchronization mechanisms prevent another request from allocating the same resource.
8. The charging session starts.
9. The database records the active allocation.
10. After charging is completed, the resource is released.
11. The database is updated with the new resource state.
12. The next waiting request can then be processed.
13. The complete activity is stored in system logs.

14. EV Charging as a Real-World Workload

EV charging is selected as the practical workload because it represents a resource-constrained environment.

14.1 Resources

The resources may include:

- Charging stations.
- Charging points.
- Available charging power.
- Charging time slots.
- Energy capacity.

14.2 Workload Characteristics

- Requests can arrive at different times.
- Multiple requests can arrive simultaneously.
- Each request may have a different charging duration.
- Some users may have different priorities.
- Charging points have limited availability.
- Resources must be released after charging.

15. Example Scenario

Consider an EV charging station having:

- 3 charging points.
- 5 EV users.
- 5 charging requests.

The system processes the requests as follows:

1. The first three eligible requests receive available charging points.
2. The remaining requests enter the waiting queue.
3. The scheduler determines the next request.
4. When a charging session completes, the charging point is released.
5. The next waiting request receives the released resource.
6. The database updates the status after every allocation and release.

16. Functional Requirements

The system should provide:

1. User registration.
2. Resource registration.
3. EV charging request creation.
4. Request validation.
5. Resource availability checking.
6. Request scheduling.
7. Resource allocation.
8. Resource reservation.
9. Resource release.
10. Transaction processing.
11. Logging.
12. Monitoring.
13. Report generation.

17. Non-Functional Requirements

- **Reliability:** Resource information should remain consistent.
- **Performance:** Requests should be processed with low unnecessary delay.
- **Scalability:** The design should support additional resources and users.
- **Security:** Only authorized users should access appropriate resources.
- **Availability:** The resource-management service should remain available during normal operation.
- **Maintainability:** Modules should be separated so that individual components can be modified.
- **Consistency:** Database state should accurately represent the current resource state.

18. Major System Modules

18.1 Module 1 – User Management

- Register users.

- Maintain user information.
- Maintain charging request history.

18.2 Module 2 – Resource Management

- Register charging stations.
- Maintain charging point status.
- Allocate resources.
- Release resources.

18.3 Module 3 – Request Management

- Create requests.
- Validate requests.
- Track request status.
- Cancel requests.

18.4 Module 4 – Scheduling

- Maintain waiting queue.
- Apply scheduling policy.
- Select next eligible request.

18.5 Module 5 – Transaction Management

- Start transactions.
- Commit successful operations.
- Roll back failed operations.
- Maintain transaction history.

18.6 Module 6 – Monitoring and Logging

- Record resource usage.
- Record allocation events.
- Record failures.
- Generate usage reports.

19. Proposed Technology Stack

Technology / Concept	Purpose
Operating System Concepts	Scheduling, synchronization and resource management.
DBMS	Storage and management of system state.
SQL	Database queries and transaction operations.
MySQL	Relational database implementation.
Java / Python / C++	Application and simulation logic.
HTML / CSS / JavaScript	User interface, if implemented.
Git / GitHub	Version control and team collaboration.

20. Development Methodology

The project will be developed in the following stages:

1. Requirement Analysis

- Identify resource-management requirements.
- Identify EV charging requirements.

2. System Design

- Design architecture.
- Design modules.

3. Database Design

- Identify entities.
- Create relationships.
- Design tables.

4. Implementation

- Implement resource management.
- Implement scheduling.
- Implement database operations.

5. Integration

- Integrate application and database.

- Integrate scheduling and resource management.

6. Testing

- Test individual modules.
- Test concurrent requests.
- Test allocation and release.

7. Evaluation

- Measure resource utilization.
- Analyze waiting time.
- Analyze allocation conflicts.

21. Testing Plan

Test Area	Purpose	Status
User Management	Check user registration and validation.	Planned
Resource Management	Check resource allocation and release.	Planned
Scheduling	Check request execution order.	Planned
Concurrency	Test simultaneous requests.	Planned
Transactions	Test commit and rollback.	Planned
Database	Check consistency of stored state.	Planned
Logging	Check system activity records.	Planned

22. Potential Challenges

1. Concurrent Resource Requests

Multiple users may request the same charging point simultaneously.

2. Race Conditions

Incorrect synchronization may result in inconsistent allocation.

3. Resource Conflicts

A charging point must not be assigned to multiple active sessions.

4. Scheduling Complexity

Different users may have different priorities and charging requirements.

5. Transaction Failure

A failed transaction should not leave the resource in an incorrect state.

6. Database Consistency

Resource status and database records must remain synchronized.

7. Scalability

The system should be able to handle an increasing number of users and resources.

23. Expected Outcomes

The project is expected to produce:

- A working prototype of a DBOS-inspired resource management platform.
- Database-backed resource state management.
- EV charging request management.
- Scheduling of competing requests.
- Safe resource allocation.
- Concurrency handling.
- Transaction-based resource management.
- Resource usage logs.
- Monitoring information.
- Demonstration of OS and DBMS integration.

24. Performance Parameters

The following parameters can be used to evaluate the system:

- **Average Waiting Time**
Time a request spends waiting before resource allocation.
- **Resource Utilization**
Percentage of time resources remain actively utilized.
- **Allocation Success Rate**
Percentage of requests successfully allocated.
- **Request Completion Time**
Time required to complete a resource request.

- **Conflict Rate**

Number of conflicting allocation attempts.

- **Transaction Failure Rate**

Number of failed transactions compared with total transactions.

25. Project Timeline

Phase	Major Activities	Status
Phase 1	Problem identification, literature review, objectives and architecture.	Completed
Phase 2	Database design, modules and initial implementation.	Upcoming
Phase 3	Integration, testing and performance evaluation.	Upcoming
Final Phase	Documentation, demonstration and final report.	Upcoming

26. Phase–1 Deliverables

The major Phase–1 deliverables are:

1. Project problem statement.
2. Project objectives.
3. Project motivation.
4. Literature review.
5. Proposed solution.
6. System scope.
7. System architecture.
8. OS concept mapping.
9. DBMS concept mapping.
10. EV charging workload definition.
11. Preliminary database design.

12. Development methodology.
13. Initial testing plan.
14. Expected outcomes.

27. Future Scope

The system can be extended in the following ways:

- Real-time EV charging station integration.
- Mobile application support.
- Real-time charging status.
- Dynamic electricity pricing.
- Smart charging.
- Renewable energy integration.
- Predictive charging demand.
- Advanced scheduling algorithms.
- Cloud deployment.
- Real-time monitoring dashboard.
- Authentication and role-based access control.
- Energy consumption analytics.
- Integration with Vehicle-to-Grid (V2G) systems.

28. Conclusion

- The proposed project combines Operating System and DBMS concepts into a unified resource management platform.
- The DBOS concept provides the foundation for database-backed system state and transactional management.
- EV charging is used as a real-world workload because it involves limited resources and concurrent user requests.
- The proposed system addresses resource allocation, scheduling, synchronization, concurrency and transaction management.
- Phase-1 establishes the problem, objectives, scope, architecture, modules, database requirements and development plan.

- Future phases will focus on implementation, integration, testing and performance evaluation.

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